Lecture 5: Probabilistic Reasoning

DD2421

André Pereira
(slides heavily adapted from Bob Sturm)

My three-lecture block

- Today: probabilistic reasoning
- Next time: learning as inference
- Next next time: learning with latent variables

My three-lecture block

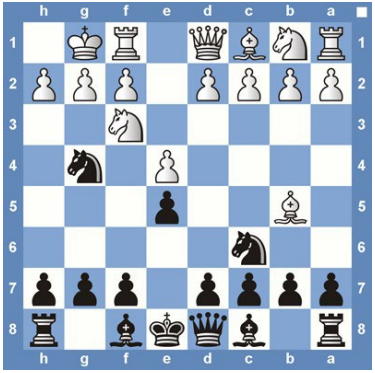
- Today: probabilistic reasoning
- Next time: learning as inference
- Next next time: learning with latent variables

Constructive alignment

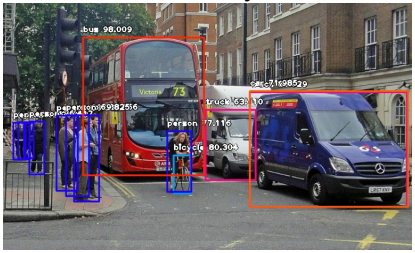
- assignments: part of lab 3 is relevant (Naïve Bayes)
- two exam problems are on probabilistic methods
- the topic is fundamental for the advanced course (DD2434)

Why Machine Learning?

AI in the 1970s



AI today



We need to deal with uncertainty!

Uncertainty

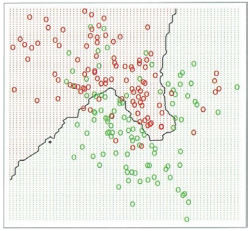
- How can we deal with the randomness we see around us?
- How can we formally describe degrees of belief?

Example: Monty Hall Problem

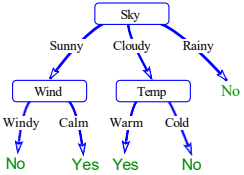


Examples of ML Methods seen so far

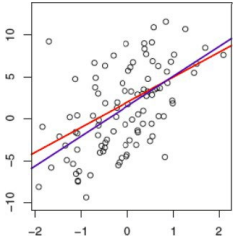
k Nearest Neighbour



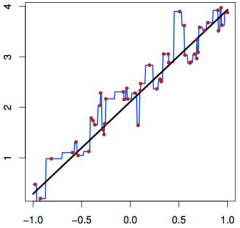
Decision Trees



Least squares Regression



k-NN Regression



Probability Theory in ML

Incorporate probabilistic thinking at all levels

- We start with incomplete knowledge (uncertainty)
- We then reduce uncertainty by incorporating observations
- We then propagate belief and update our estimations.

Probabilistic machine learning provides
something like a unified theory for ML!

Advantages of Probability Based Methods

- **Interpretability!** Can be more transparent and mathematically rigorous than other ML methods. (But humans can have problems interpreting probability, as we will see later today.)
- **Transparency!** Assumptions can be made more explicit than in other methods (we will see this when we look at probabilistic linear regression next time).
- **Efficiency!** Can work in regimes that are data poor.
- **Flexibility!** Easy to merge different parts of a complex system and to update current knowledge with new observations.
- **Encompassing!** Aspects of learning and inference can be cast under the same theory.

Disadvantages of Probability Based Methods

- **Complicated ...** Often hard to derive closed solutions. Need to resort to computation and heuristic approximations.
- **Scalability ...** Not computationally scalable to large datasets (but many argue that the need for large data sets is a disadvantage).

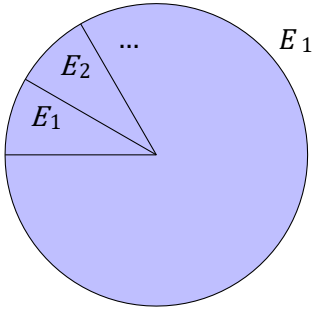
Outline

1. Probability Theory Reminder
   - Axioms and Properties
   - Common Distributions
   - Expectation
2. Probabilistic Machine Learning
   - Supervised Learning, General Definition
   - Regression
   - Classification

Axiomatic definition of probabilities (Kolmogorov)

Given a *sample space* Ω of all possible *outcomes*, and an *event* E, or set of outcomes from Ω (possibly empty), then:

1. $P(E) \geq 0$ for all $E \subseteq \Omega$
2. If $E = \Omega$ then $P(\Omega) = 1$ (sure event) $\Rightarrow 0 \leq P(E) \leq 1$
3. If $E_1, E_2, \ldots$ is a countable sequence of pairwise disjoint events, then

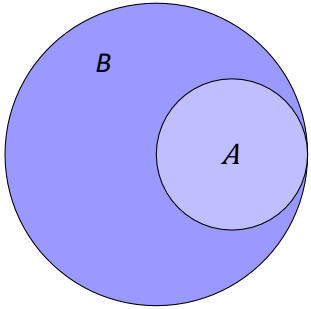


$E_1 \cup E_2 \cup \ldots$

$$P(E_1 \cup E_2 \cup \cdots) = \sum_i P(E_i)$$

Consequences

1. Monotonicity: $P(A) \leq P(B)$ if $A \subseteq B$



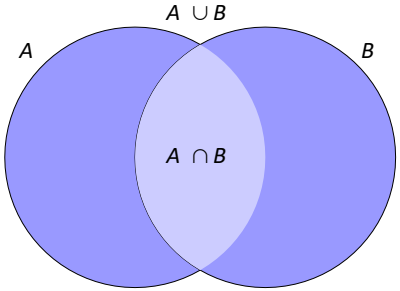
Example (die): $A = \{3\}$, $B = \{\text{odd}\}$

2. Empty set $\emptyset$: $P(\emptyset) = 0$

Example (die): $P(A \cap B)$ where $A = \{\text{odd}\}$, $B = \{\text{even}\}$

More Consequences: Addition

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Example:

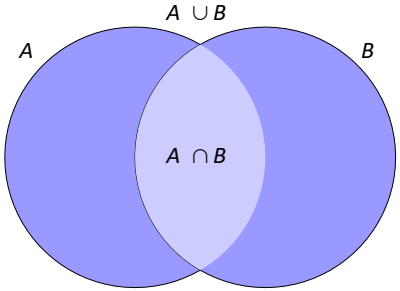
$A = \{1, 3, 5\}$, $P(A) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$

$B = \{5, 6\}$, $P(B) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

$A \cap B =$

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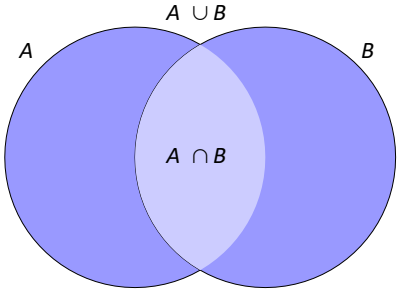
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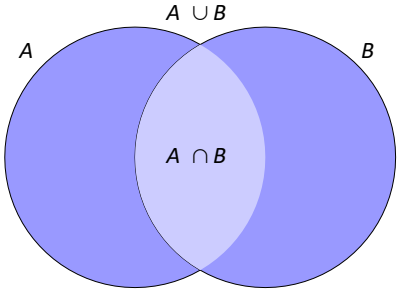
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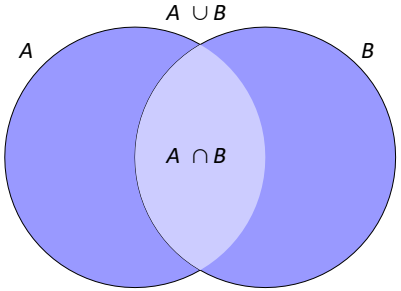


Example:

$$\begin{aligned}
A &= \{1, 3, 5\}, & P(A) &= \tfrac{1}{6} + \tfrac{1}{6} + \tfrac{1}{6} = \tfrac{1}{2} \\
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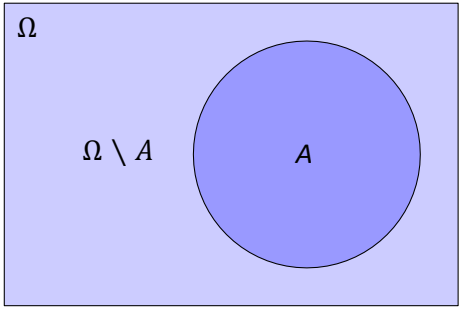


Example:

$$\begin{array}{llll} A = \{1,3,5\}, & P(A) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2} \\ B = \{5,6\}, & P(B) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3} \\ A \cap B = \{5\} & P(A \cap B) = \frac{1}{6} \\ A \cup B = \{1,3,5,6\} & P(A \cup B) = \frac{1}{2} + \frac{1}{3} - \frac{1}{6} = \frac{2}{3} \end{array}$$

More Consequences: Negation

$$P\left(\overline{A}\right) = P\left(\Omega \setminus A\right) = 1 - P\left(A\right)$$



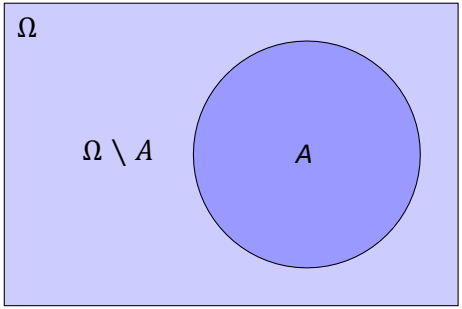
Example:

$A = \{1, 2\}, \qquad P(A) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

$\bar{A} =$

More Consequences: Negation

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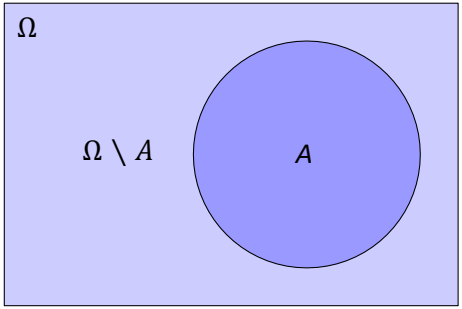


Example:

$$A = \{1, 2\}, \quad P(A) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$
$$\bar{A} = \{3, 4, 5, 6\}, \quad P(\bar{A}) =$$

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Random (Stochastic) Variables

- Sample spaces (set of all possible outcomes) are not numbers, e.g., die rolls, heads/tails, playing cards, words, etc.
- We need to “convert” outcomes to numbers.
- Let’s map the set of outcomes to a set of numbers!

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- Let's map the set of outcomes to a set of numbers!

A *random variable (r.v.)* is neither random nor a variable. It's a function that does exactly what we need, e.g.,

$$X : \Omega \to \mathbb{R}$$

Probability distributions

The *probability distribution function (PDF)* of a r.v. maps its range to positive numbers

$$p(x): X \rightarrow \mathbb{R}_{\geq 0}$$

- $p(x)$ describes how probability *density* is distributed over the range of X, e.g., the probability of $0 \leq X \leq 1$ is given by:

$$P[0 \leq X \leq 1] = \int_0^1 \mathrm{p}(x)\ dx$$

- We say that X is distributed according to $p(x)$", written informally as:

$$X \sim \mathrm{p}(x)$$

Examples

Student in HT19 machine learning course hands in exam, which is then graded.

Ω – All possible graded exams of all students in class

$X : \Omega \to \mathbb{R}$ – Random variable mapping exam outcome to score

$X \sim \mathcal{N}(32.5, 75.5)$ – Gaussian distribution, $\mu = 32.5$, $\sigma^2 = 75.5$

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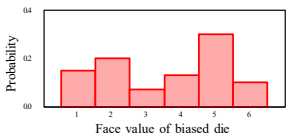
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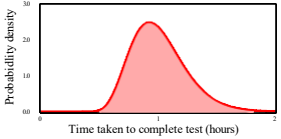
Snälla: Work hard so that μ is much higher!

Types of Random Variables

- The range of a **discrete random variable** is a countable set, e.g., $\{1, 2, \ldots, 6\}$.
- The range of a **continuous random variable** is an uncountable set, e.g., the reals R, or $[0, 1)$.



discrete pdf



continuous pdf

Figures taken from Computer Vision: models, learning and inference by Simon Prince.

Examples of Random Variables



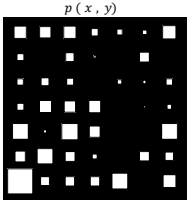
- Discrete events: either 1, 2, 3, 4, 5, or 6.
- Probability **mass** function (PMF) $P[X = d] = p(d)$
- $P[X = d] = 1/6, d \in \{1, 2, 3, 4, 5, 6\}$ (fair die)



- Any real number (theoretically infinite)
- Probability **density** function (PDF) $p(x)$
- $P[t_1 < X < t_2]$ involves integrating $p(x)$
- $P[X = t_1] = 0$ but often $p(t_1) \neq 0$

Joint Probabilities

- Consider two random variables X and Y.
- Observe multiple paired instances, $(x, y)_k$. Some paired outcomes will occur more frequently.
- This information is encoded in the *joint* probability density function $p(x, y)$.



→ **discrete joint distribution (**PMF-like representation)

Figure from **Computer Vision: models, learning and inference** by Simon Prince.

Marginalization

The distribution of any single variable can be recovered from a joint distribution by summing for the discrete case

$$\mathrm{p}(x) = \sum_{y} \mathrm{p}(x, y)$$

and integrating for the continuous case

$$\mathrm{p}(x) = \int \mathrm{p}(x, y)\ dy$$

Marginalization (cont.)

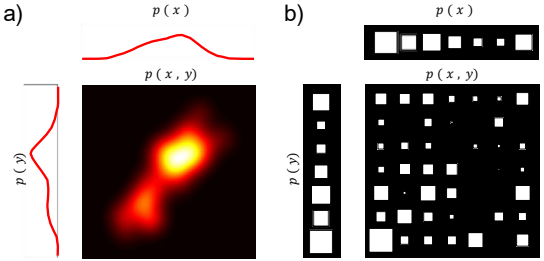


Figure from **Computer Vision: models, learning and inference** by Simon Prince.

Conditional Probabilities

$$P(A|B)$$

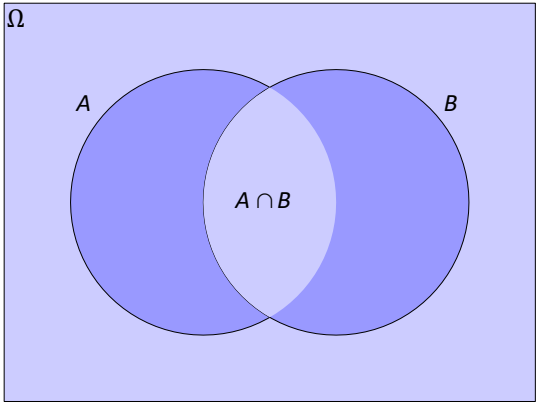
The probability of event *A* *given* that event *B* has happened

Note: This is not the probability that event *A* *and B* will happen



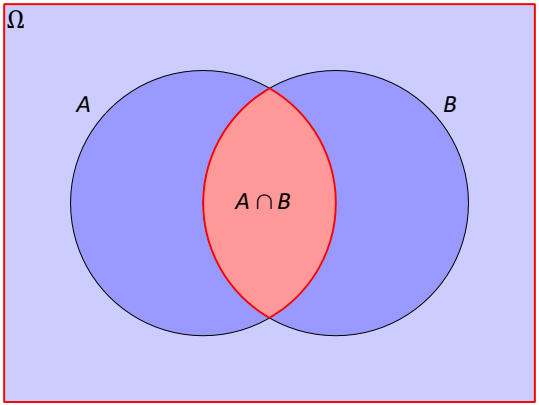
Conditional Probabilities

$$P(A|B) \neq P(A \cap B)$$



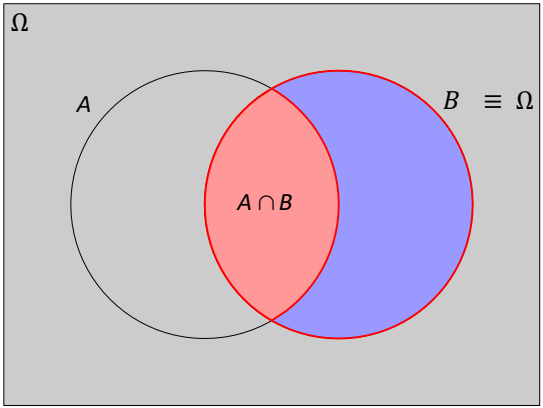
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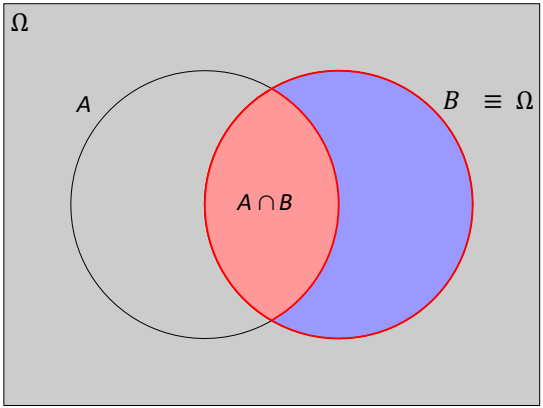
Conditional Probabilities

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Conditional Probabilities

$$P(A|B) = \frac{\mathrm{P(A \cap B)}}{\mathrm{P(B)}}$$



Conditional Probability (Random Variables)

- The *conditional probability distribution* of X , given that Y takes value y, tells us how likely different values of x are, assuming $Y = y$.
- The conditional probability can be found from the joint distribution function $p(x, y)$:

$$p(x \mid Y = y) = \frac{p(x, Y = y)}{p(Y = y)} = \frac{p(x, Y = y)}{\int_x p(x, Y = y)\, dx}$$

- Extract an appropriate slice and then normalize it.

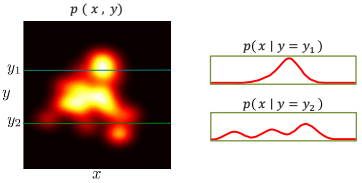


Figure from **Computer Vision: models, learning and inference** by Simon Prince.

Independence

- Two events are *independent* if the joint distribution can be factorized: $P(A \cap B) = P(A)P(B)$
- this means that:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A)$$

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Knowing that B happened tells us nothing new about A happening

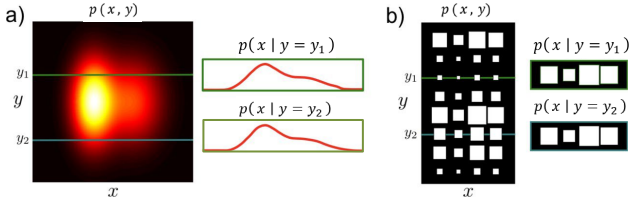


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Bayes' Theorem

if

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

then

$$P(A \cap B) = P(A \mid B)\, P(B)$$

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Bayes' Theorem

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$$P(A \cap B) = P(A \mid B)\,P(B) = P(B \mid A)\,P(A)$$

and

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}$$

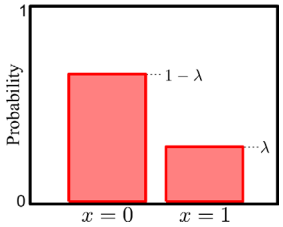
Memorize this and its derivation!

Bernoulli: binary variables

- Domain: binary variables ($x \in \{0, 1\}$)
- Parameters: $\lambda = P[x = 1], \lambda \in [0, 1]$

$Then\ P[x = 0] = 1 - \lambda$, and

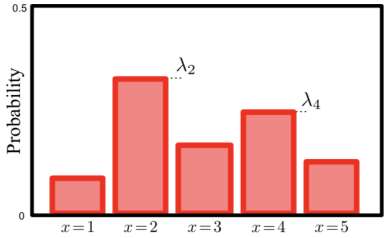
$$p(x) = \lambda^x (1 - \lambda)^{1-x} = \begin{cases} \lambda, & \text{if } x = 1 \\ 1 - \lambda, & \text{if } x = 0 \end{cases}$$



Categorical

- Domain: discrete variables ($x \in \{x_1, \ldots, x_K\}$)
- Parameters: $\lambda = [\lambda_1, \ldots, \lambda_K]$
- with $\lambda_k \in [0, 1]$ and $\sum_{k=1}^{K} \lambda_k = 1$

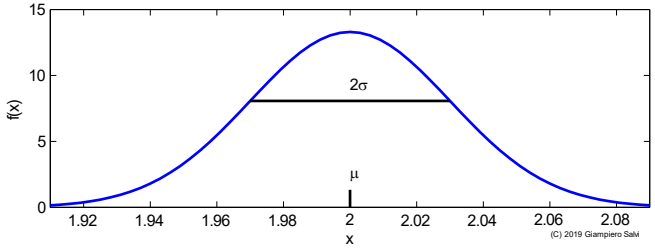
$$P[x_k = 1] = \lambda_k$$



Gaussian distributions: One-dimensional

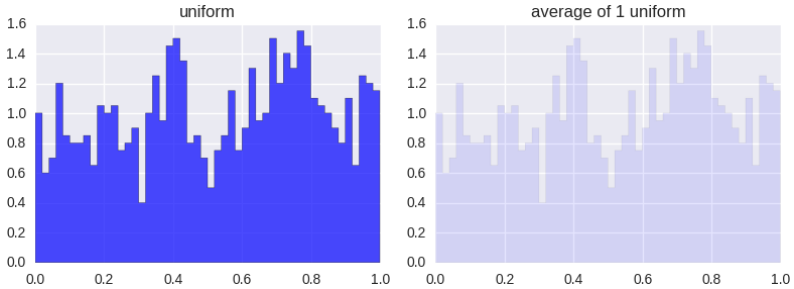
- Univariate normal distribution
- Domain: real numbers ($x \in \mathbb{R}$)

$$\mathrm{p}(x; \mu, \sigma^2) = \mathcal{N}(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$



Why Gaussian: Central Limit Theorem

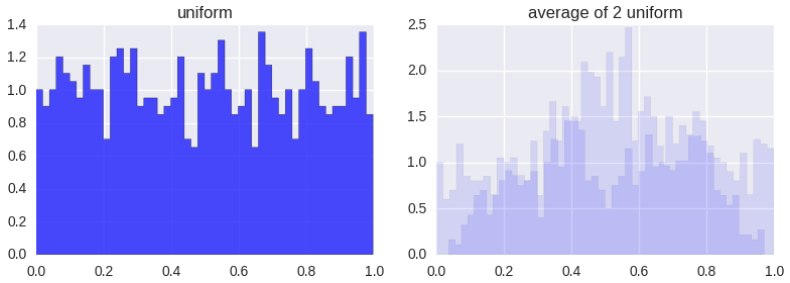
The distribution of a linear combination of a large number of independent, identically distributed (iid) variables will tend to normal, regardless of the underlying distribution.[1]



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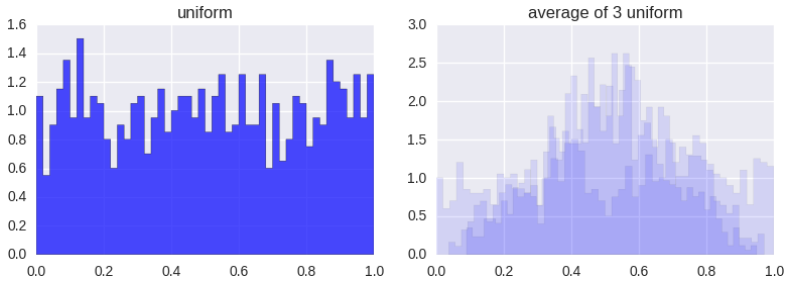
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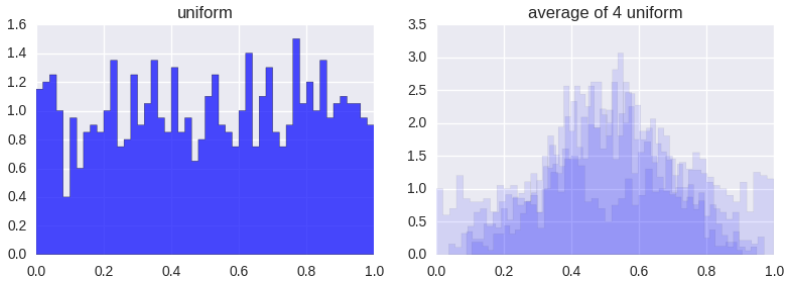
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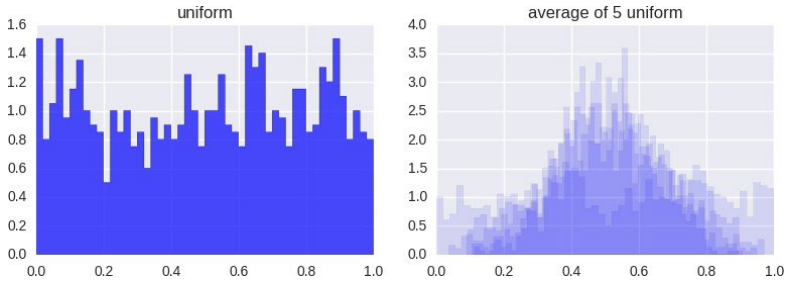
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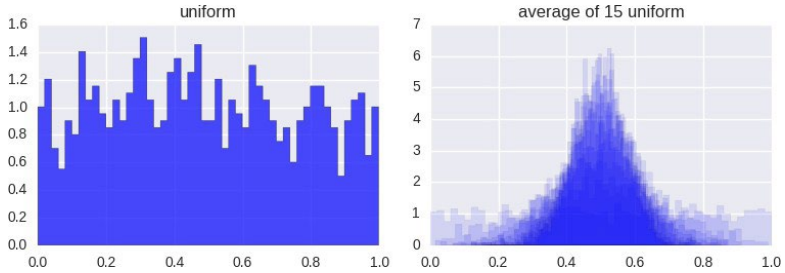
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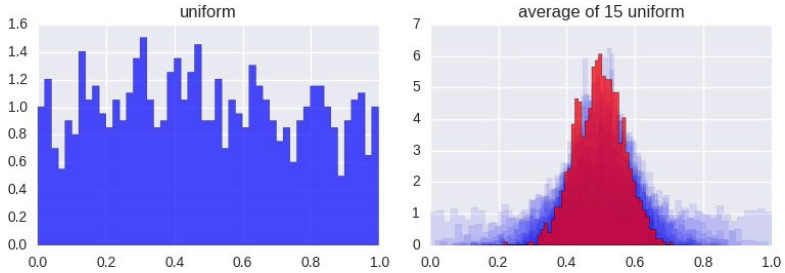
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Why Gaussian: Central Limit Theorem

Galton Board (Sir Francis Galton, 1822-1911)



Gaussian distributions: D Dimensions

- Also know as the *multivariate* normal distribution
- Domain: real numbers ($\boldsymbol{x} \in \mathbb{R}^D$)

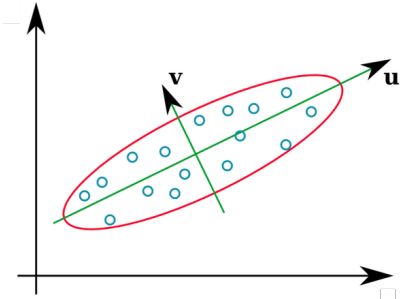
$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_D \end{bmatrix} \qquad \boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \dots \\ \mu_D \end{bmatrix} \qquad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \dots & \sigma_{1\mathrm{D}} \\ \sigma_{21} & \sigma_2^2 & \dots & \\ \dots & & & \\ \sigma_{\mathrm{D}1} & \dots & & \sigma_{\mathrm{D}}^2 \end{bmatrix}$$

$$p(\boldsymbol{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{\mathrm{D}}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} exp\left[-\frac{1}{2} (\boldsymbol{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-}1 (\boldsymbol{x} - \boldsymbol{\mu}) \right]$$

Gaussian distributions

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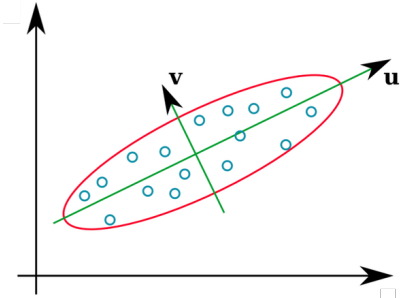
Eigenvalue decomposition of the covariance matrix: $\Sigma = Q\,\Lambda\,Q^T$



Gaussian distributions

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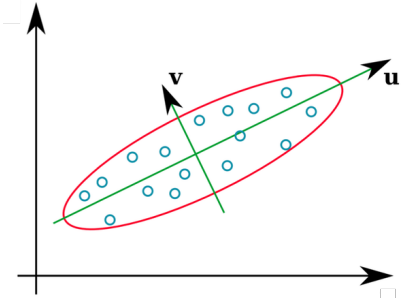
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Gaussian distributions

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Eigenvalue decomposition of the covariance matrix: $\Sigma = Q\,\Lambda\,Q^T$



Expected value

$$\mathbb{E}[X] = \mu_X = \int x \; p(x) \; dx$$

$$\mathbb{E}[\mathbf{x}] = \mu_{\mathbf{x}} = \int \mathbf{x} \; p(\mathbf{x}) \; d\mathbf{x}$$

- The “center of gravity” of a distribution
- Sampled expected value (mean)

$$\overline{\mu_X} = \frac{1}{N} \sum_{i=1}^{N} x_i$$

Variance

$$\mathrm{Var}[X] = \sigma_X^2 = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

- The "spread" of a distribution
- Unbiased sample variance (using N − 1)

$$\widehat{\sigma_X^2} = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2$$

Covariance

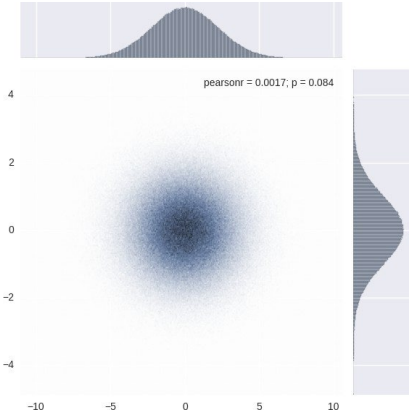
$$\sigma_{X,Y} = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

$$\Sigma_{\mathbf{x}} = \mathbb{E}[(\mathbf{x} - \mathbb{E}[\mathbf{x}])(\mathbf{x} - \mathbb{E}[\mathbf{x}])^T]$$

- Shows how two variables vary *together*
- Unbiased sample co-variance

$$\widehat{\sigma_{X,Y}} = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})$$

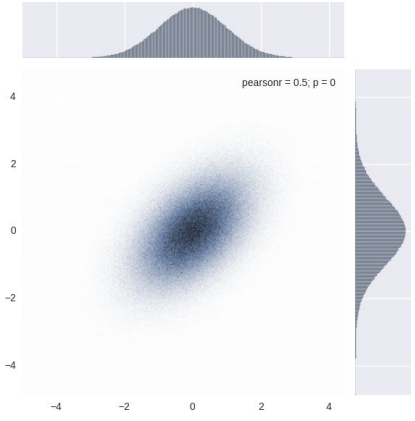
Examples



2

[2]Script by C.E. Ek

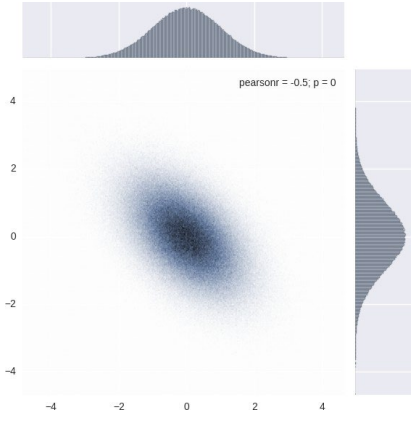
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2

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Examples

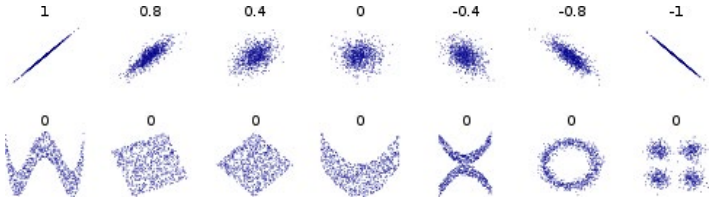


2

[2]Script by C.E. Ek

Covariance and Independence

- covariance shows "linear" dependency
- dependent variables may have zero covariance
- in some distributions zero covariance is equivalent to independence (example?)



[3]

[3]Figure adapted from Wikipedia

Outline

1. Probability Theory Reminder
   - Axioms and Properties
   - Common Distributions
   - Expectation
2. Probabilistic Machine Learning
   - Supervised Learning, General Definition
   - Regression
   - Classification

General ML problem (supervised learning)

Data:

$$\{(\boldsymbol{x}, y)1, (\boldsymbol{x}, y)2, \ldots, (\boldsymbol{x}, y)_n\}$$

Where **x** are features (independent variables), and y is the answer (dependent variable).

- if Y is discrete: *classification*
- if Y is continuous: *regression*

General ML problem (supervised learning)

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Where **x** are features (independent variables), and y is the answer (dependent variable).

- if Y is discrete: *classification*
- if Y is continuous: *regression*

Learning: understand how variables are related

- estimate joint probability $p(\mathbf{x}, y)$ from observations $\{(\mathbf{x}, y)_j\}$

Inference: predict an answer given an observation

- estimate the conditional probability $p(y|\mathbf{X} = \mathbf{x})$

Bayes' Rule

$$\mathrm{p}(\mathrm{y} \mid \mathbf{X} = \mathbf{x}) = \frac{\mathrm{p}(\mathbf{x} \mid \mathrm{Y} = \mathrm{y})\,\mathrm{p}(\mathrm{Y} = \mathrm{y})}{\mathrm{p}(\mathbf{X} = \mathbf{x})}$$

- $\mathrm{p}(\mathbf{x} \mid \mathrm{Y} = \mathrm{y})$ → Likelihood represents the probability density of observing data $\mathbf{x}$ given the *hypothesis* $Y = y$.
- $\mathrm{p}(Y = y)$ → Prior represents the knowledge about Y before any observation.
- $\mathrm{p}(y \mid \boldsymbol{X} = \boldsymbol{x})$ → **Posterior** represents the probability density of hypothesis y given observation $\mathbf{X} = \mathbf{x}$.
- $\mathrm{p}(\boldsymbol{X} = \boldsymbol{x})$ → Evidence ensures the posterior is a proper probability distribution.

$$\mathrm{p}(\mathbf{X} = \mathbf{x}) = \begin{cases} \sum_{\mathrm{y}} \mathrm{p}(\mathbf{x} \mid \mathrm{Y} = \mathrm{y})\,\mathrm{p}(\mathrm{Y} = \mathrm{y}) & \text{classification} \\ \int_{\mathrm{y}} \mathrm{p}(\mathbf{x} \mid \mathrm{Y} = \mathrm{y})\,\mathrm{p}(\mathrm{Y} = \mathrm{y})\,\mathrm{dy} & \text{regression} \end{cases}$$

Probabilistic Regression

$$(X, Y) \sim N(\mu, \Sigma)$$

Regression via conditional probability:

- Find joint distribution of X and Y : $\mathrm{p}(x, y)$
- Compute posterior of Y :
$$p(y|X = x) = \mathrm{p}(\boldsymbol{x}, y)/\mathrm{p}(X = x)$$
- Compute conditional expectation: $\mathbb{E}[Y|X = x]$

Explicit regression model:

- Define a deterministic model $y = f(\boldsymbol{x}) + \epsilon$
- Describe probability distribution of the error $\epsilon = y - f(\boldsymbol{x})$
- Estimate parameters in $f(\boldsymbol{x})$

Example: Bivariate Normal distribution

Define joint probability distribution function

$$(X, Y) \sim \mathcal{N}(\mu, \sigma^2)$$

Where:

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{bmatrix}$$

and ρ is the *correlation coefficient*. The posterior distribution is Normal:

$$p(y \mid X = x; \mu, \Sigma) = \mathcal{N}(\mu_{Y|X=x}, \sigma^2_{Y|X=x})$$

where:

$$\mu_{Y|X=x} = \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X) = w_0 + w_1 x$$

$$\sigma^2_{Y|X=x} = (1-\rho^2)\sigma_Y^2 \text{ (constant wrt. } x\text{)}$$

Explicit Regression Model

Model (deterministic):

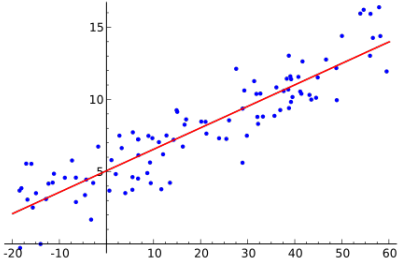
$$y = \boldsymbol{w}^T\mathbf{x} + \boldsymbol{\epsilon}$$

But now:

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mu, \sigma^2)$$

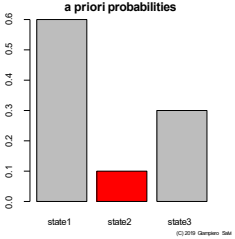
Therefore:

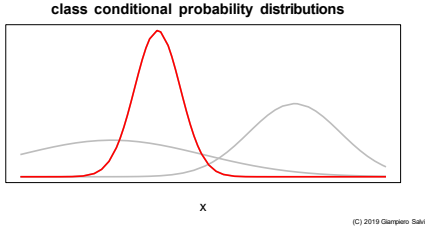
$$Y \sim \mathcal{N}\left(\mu_Y(\mathbf{x}), \sigma_Y^2(\mathbf{x})\right) = \mathcal{N}(\boldsymbol{w}^T\mathbf{x}, \sigma^2)$$



The Probabilistic Model of Classification

- One of K states is selected with *a priori* probability
$$\mathrm{p}(Y = y)$$
- Given state $Y = y$, an observation $\boldsymbol{X} \sim \mathrm{p}(\boldsymbol{x}|Y = y)$





Problem

If I observe a new $\hat{\mathbf{x}}$, and I know $\mathrm{p}(Y = y)$ and $p(\boldsymbol{x}|Y = y)$ for each of K classes, what can I say about $p(y|\boldsymbol{X} = \hat{\mathbf{x}})$?

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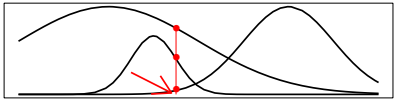
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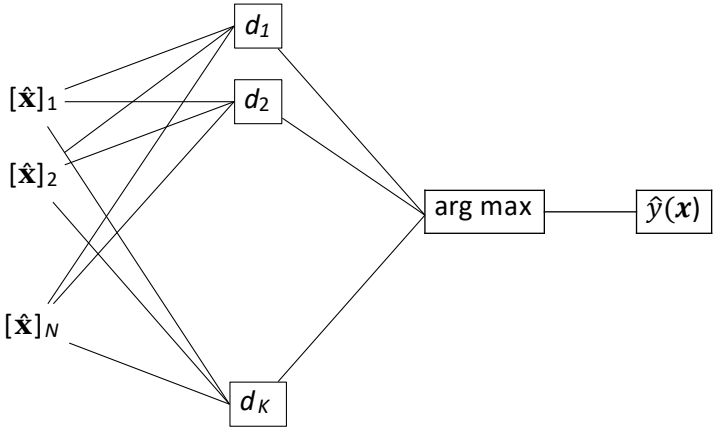
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$$\begin{aligned}\mathrm{p}(\mathrm{X}=\hat{\mathbf{x}}) = {} & \mathrm{p}(\mathrm{X}=\hat{\mathbf{x}} \mid \mathrm{y} = \boldsymbol{y_1})\,\mathrm{p}(\mathrm{Y} = y_1) + \\ & \mathrm{p}(\mathrm{X}=\hat{\mathbf{x}} \mid \mathrm{y} = \boldsymbol{y}_2)\,\mathrm{p}(\mathrm{Y} = y_2) + \\ & \mathrm{p}(\mathrm{X}=\hat{\mathbf{x}} \mid \mathrm{y} = \boldsymbol{y}_3)\,\mathrm{p}(\mathrm{Y} = y_3)\end{aligned}$$



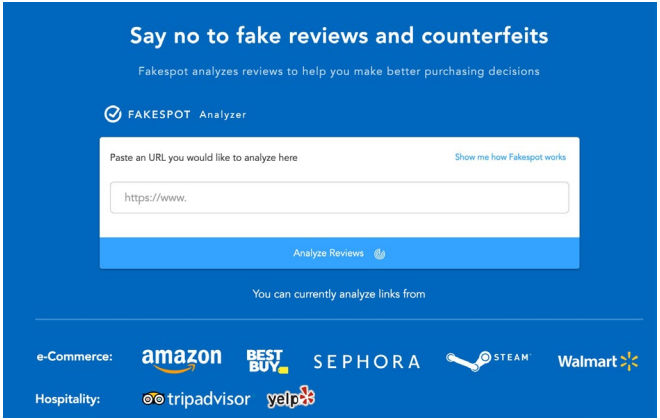
Classifiers: Discriminant Functions

$$d_k(\hat{\mathbf{x}}) = \mathrm{p}(\hat{\mathbf{x}} \mid Y = y_k)\, \mathrm{p}(Y = y_k)$$



Example: Is this online product review real or fake?

Task: Determine whether an online product review is real or fake given its length in characters.



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Information given:

- Somehow, someone on the inside gave us a dataset of online product reviews labeled as real and fake. (Don't ask too many questions.)
- We can thus estimate the priors
  $p[Real] = p[G = +1] = 0.25, \quad p[Fake] = p[G = -1] = 0.75$
- And we can estimate the likelihood distributions $p(x \mid G = +1)$ and $p(\text{x} \mid G = -1)$ (after we make a decision about the distribution)

Example: Is this online product review real or fake?

Task: Determine whether an online product review is real or fake given its length in characters. $\Rightarrow$ Find $\mathrm{p}(\mathrm{g} | \mathrm{X} = \mathrm{x})$.

Solution: Apply Bayes' Rule!

$$\mathrm{p}(\mathrm{g} \mid \mathrm{X} = \mathrm{x}) = \frac{\mathrm{p}(\mathrm{X} = \mathrm{x} \mid \mathrm{G} = \mathrm{g})\, \mathrm{p}(\mathrm{G} = \mathrm{g})}{\mathrm{p}(\mathrm{X} = \mathrm{x})}$$

$$= \frac{\mathrm{p}(\mathrm{X} = \mathrm{x} \mid \mathrm{G} = \mathrm{g})\, \mathrm{p}(\mathrm{G} = \mathrm{g})}{\mathrm{p}(\mathrm{X} = \mathrm{x} \mid \mathrm{G} = +1)\, \mathrm{p}(\mathrm{G} = +1) + \mathrm{p}(\mathrm{X} = \mathrm{x} \mid \mathrm{G} = -1)\, \mathrm{p}(\mathrm{G} = -1)}$$

We can now calculate $p(G = +1 \mid X = x)$

Selecting the most probable hypothesis

- **Maximum A Posteriori (*MAP*):**

  Choose hypothesis from Y with highest probability given observed data $\mathbf{x}$:

$$\begin{aligned} y_{MAP}(\mathrm{x}) &= \arg\max_{y \in \mathrm{Y}} \; p(Y = y \mid \mathbf{X} = \mathbf{x}) \\ &= \arg\max_{y \in \mathcal{Y}} \; \frac{p(\mathbf{X} = \mathbf{x} \mid Y = y)\, p(Y = y)}{p(\mathbf{X} = \mathbf{x})} \\ &= \arg\max_{y \in \mathcal{Y}} \; p(\mathbf{X} = \mathbf{x} \mid Y = y)\, p(Y = y) \end{aligned}$$

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- **Maximum Likelihood (*ML*):**

  If we do not know prior distribution, then choose hypothesis with highest likelihood of generating the observed data:

$$y_{MLE}(\mathrm{x}) = \arg\max_{y \in \mathcal{Y}} \; p(\mathbf{X} = \mathbf{x} \mid Y = y)$$

Summary

Today:

1. Probability Theory Reminder
   - Axioms and Properties
   - Common Distributions
   - Expectation
2. Probabilistic Machine Learning
   - Supervised Learning, General
   - Definition Regression
   - Classification

Next time: How to *fit* probability models to data (estimating parameters with the maximum likelihood optimality criterion)